

NATIONAL TRANSPORTATION SAFETY BOARD

Office of Research and Engineering
Materials Laboratory Division
Washington, D.C. 20594



December 7, 2006

MATERIALS LABORATORY STUDY

Report No. 06-090

A. ACCIDENT

Place : Boston, Massachusetts
Date : July 10, 2006
Vehicle : Connector Tunnel to Ted Williams Tunnel
NTSB No. : HWY06MH024
Investigator : Mark Bagnard

B. COMPONENTS EXAMINED

Simulated anchor assemblies

C. DETAILS OF THE STUDY

Test Setup

On October 10, and October 31, 2006, torque versus axial load tests were performed on simulated anchor assemblies at the Federal Highway Administration's Turner Fairbank Highway Research Center. A 6-foot long 316 stainless steel threaded rod¹ was extended through a Strainert flat load cell (model number is 08905-7, load capacity 100,000 pounds) and the strong (reinforced concrete) floor (approximately 3 feet thick) of a section of the facility. A thick plate washer and a nut threaded onto the upper end of the rod prevented the rod from dropping through the floor and reacted load into the load cell. The threaded rod extended through the underside of the floor several inches. Using a Proto 6016 (30-150 foot-pounds) torque wrench, torque was applied to a nut attached at the lower end of the rod under various test configurations and torque values described below, and axial load was measured as summarized in the table at the end of this report.

In all tests, the load cell was reset to zero with the threaded rod in place. Also, all tests utilized a square 0.5-inch thick steel plate that spanned the 3-inch hole in the strong floor, a 0.375 thick octagonal steel washer with a 0.75 inch diameter hole, and a new galvanized 0.1-inch thick steel washer (outside diameter 1.8 inches, inside diameter 0.7 inch) directly on the underside of the floor. Figure 1 shows the components on the underside of the floor prior to test 3, with the various components labeled. During the

¹ 5/8 inch diameter with 11 threads per inch, the same as the anchors for the suspended ceiling in the tunnel.

testing to each torque level, the maximum load exerted on the load cell was captured. In addition, the load on the load cell was recorded just after each target torque level was reached, and 1 to 2 minutes later to measure any decrease (relaxation) in the load. At the end of each test, load was removed by unscrewing the nut at the lower end of the rod. Between tests, the threaded rod was displaced downward a slight amount so that virgin threads were used for each test.

Test 1. A new zinc-plated nut was put on the threaded rod, and the nut was torqued to 125 foot-pounds.

Test 2. The lock washer from anchor 3-3² from the failure location in the tunnel and the stainless steel nut from anchor 3-3 were put on the threaded rod and the nut was torqued to 30 foot-pounds, 50 foot-pounds, 75 foot-pounds, 100 foot-pounds, and 125 foot-pounds.

Test 3. The sealing (rubber/metal) washer from anchor 5-2² from the failure location in the tunnel, the lock washer from anchor 5-2, and the stainless steel nut from anchor 5-2 were put on the threaded rod and the nut was torqued to 30 foot-pounds, 50 foot-pounds, 75 foot-pounds, 100 foot-pounds, and 125 foot-pounds.

Test 4. A new zinc-plated nut was put on the threaded rod and the nut was torqued to 30 foot-pounds, 50 foot-pounds, 75 foot-pounds, 100 foot-pounds, and 125 foot-pounds.

Test 5. The sealing (rubber/metal) washer from anchor 6-1 from the failure location in the tunnel and a new galvanized nut (ASTM A194 grade 2H, hot dipped galvanized steel) were put on the threaded rod and the nut was torqued to 30 foot-pounds, 50 foot-pounds, 75 foot-pounds, 100 foot-pounds, and 125 foot-pounds.

Discussion / Comments

The torque wrench used in the test measured 20 inches from the center of the socket to the end of the wrench. Application of the 125 foot-pounds overhead required near maximum effort, and jerking of the handle (sudden application of load) was used in order to reach this torque level.

Lock washers from accident hardware were used in tests 2 and 3. Removal of the nut following completion of the test resulted in the creation of an in-plane gap between the points of the washer in addition to the initial out-of-plane offset.

The sealing (rubber/metal) washer from accident hardware had been somewhat deformed before the beginning of torque test 3. It was noted that during application of the 125 foot-pounds of torque, the rubber portion of the sealing washer was substantially deformed and was squeezed outward. Upon disassembly, it was observed that the rubber

² See Materials Laboratory Factual Report No. 06-091 for a description of the anchors and hardware from the failure location.

portion had separated from the metal portion, and significant additional shredding and damage had occurred to the rubber portion of this washer. Figure 2 shows the sealing washer from the 5-2 anchor after the torque test in comparison to the sealing washer from the 5-1 anchor, which was not used in a torque test. Similar damage occurred to the sealing washer from the 6-1 anchor used in test 5.

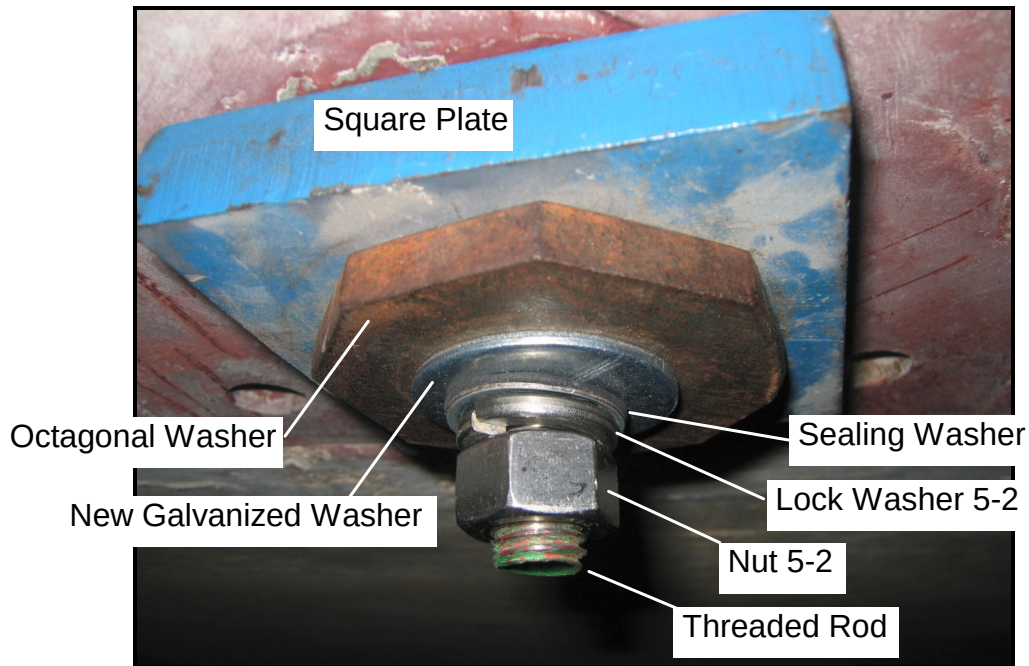
Results of Load Measurements

The following table shows the recorded results from the testing.

	Torque (foot-pounds)	Maximum Load (pounds)	Relaxed Load-Immediate	Relaxed Load-1 to 2 Minutes
Test 1	125	9,430	9,260	9,170
Test 2	30	2,760	N/R*	N/R*
	50	4,640	4,510	N/R*
	75	6,770	6,690	6,660
	100	8,370	8,240	8,210
	125	9,790	9,730	9,670
Test 3	30	2,730	2,500	2,440
	50	3,960	3,920	3,890
	75	5,750	5,590	5,560
	100	7,020	6,980	6,960
	125	8,360	8,280	8,250
Test 4	30	3,310	3,260	3,250
	50	5,230	5,190	5,160
	75	7,970	7,900	7,870
	100	9,680	9,550	9,500
	125	11,250	11,050	10,980
Test 5	30	2,710	2,610	2,550
	50	4,810	4,740	4,670
	75	7,740	7,350	7,280
	100	9,050	8,980	8,900
	125	10,840	10,790	10,690

*Not Recorded

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ImageNo:0610A00248, Project No:2006100008

Figure 1. Components on the underside of the floor prior to test 3.



ImageNo: 0610A00247, Project No:2006100008

Figure 2. Comparison photographs of the sealing washer from anchor 5-2 (on the right) after torque testing and the sealing washer from anchor 5-3 (on the left), which was not utilized in the testing.